

# Managing the Big Data Deluge in Enterprises with Hadoop

Data is the new oil of the modern digital economy. Across the globe, enterprises are generating data at a pace like never before. This data is in different formats and from various sources – millions of financial transactions, sensor data from IoT devices, industrial processes and automobiles, and from our ubiquitous smartphones, fueled by our addiction to social media. So let's have a look at why Big Data is so important and why Hadoop is a life saver when it comes to the analysis of this data.



**T**here is a tremendous amount of data that is getting generated. Activities like social media interactions, surfing the Internet, user communications, online shopping, sharing and Internet searches produce massive streams of data. These data sets are so large or complex that traditional data processing application software are inadequate to deal with them. To quantify the data that is getting generated and to put things in perspective, let's do a quick run-down of the vital statistics of the modern digital world:

- There were 5 exabytes of information created between the dawn of civilisation through 2003, but that much information is now created every two days.
- By 2020, one million devices are expected to come online every hour.
- The Weather Channel receives more than 18 million weather forecast requests every minute.
- Data is expected to grow to 44ZB by 2020.
- By 2020, the average age of an S&P 500 enterprise is expected to be about 12 years. This was 60 years in 1960.
- More than 90 per cent of the current world data was generated in the last two years.
- IoT devices are expected to grow to 200 billion by 2020.
- Over 4 million videos are viewed on YouTube every minute.
- On an average, Google processes

over 40K searches every second. This translates to a whopping 3.5B searches every day.

- Over 65 billion messages are sent via the WhatsApp mobile app and Web client every day.
- More than 300 million pictures are uploaded to Facebook every day.
- Netflix users stream around 70K hours of video every minute.
- Social media is the biggest generator of data. Every minute, around 500K tweets are sent, over 150K Skype calls are made, over 100 million spam emails are sent, and around 600K pictures are shared on Snapchat.

The data that is getting generated has certain specific characteristics which qualifies them as Big Data. These characteristics are called the 5 V's of Big Data

## The Five Vs of Big Data

All that gets classified as Big Data shares certain specific characteristics that are called the Five Vs, which include the following.

**Volume:** This refers to the humongous amount of data that is getting generated every day, every minute and every second. Data that is far more voluminous than traditional relational database systems can manage constitutes Big Data. These huge data sets warrant the use of a different computing paradigm to store and manage them. Big Data leverages the power of distributed computing to extract actionable insights out of huge data sets.